UNIVERSITÉ DE GENÈVE
FACULTY OF SCIENCE
Department of Informatics

SIP stochastic information processing

Modern steganography

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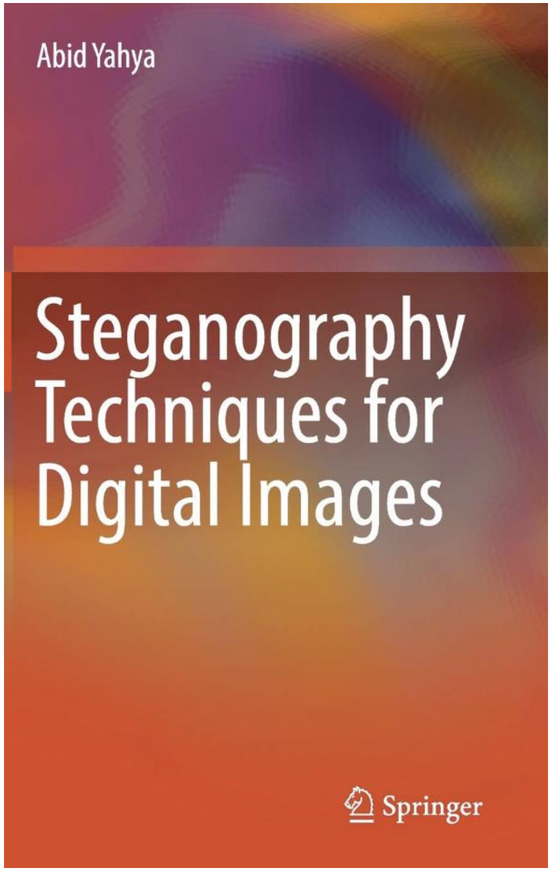
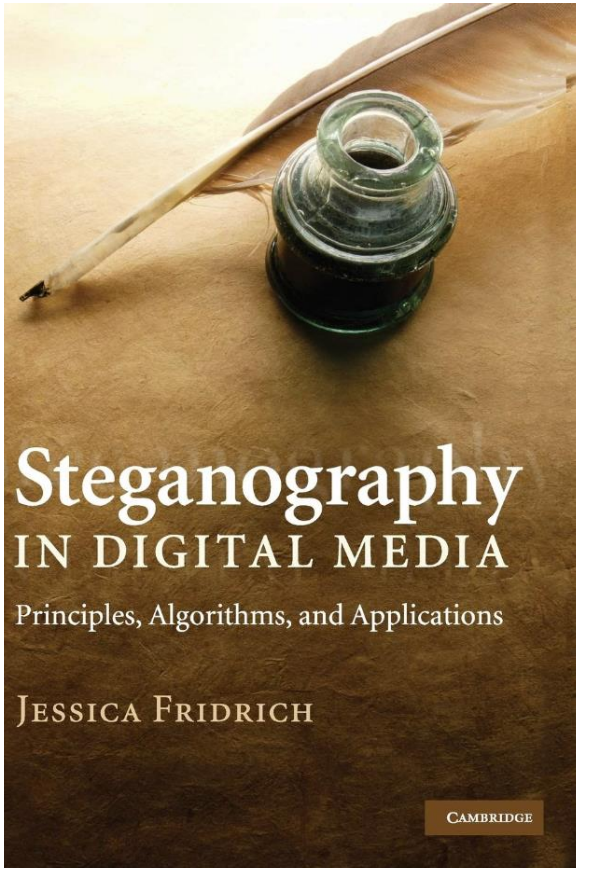
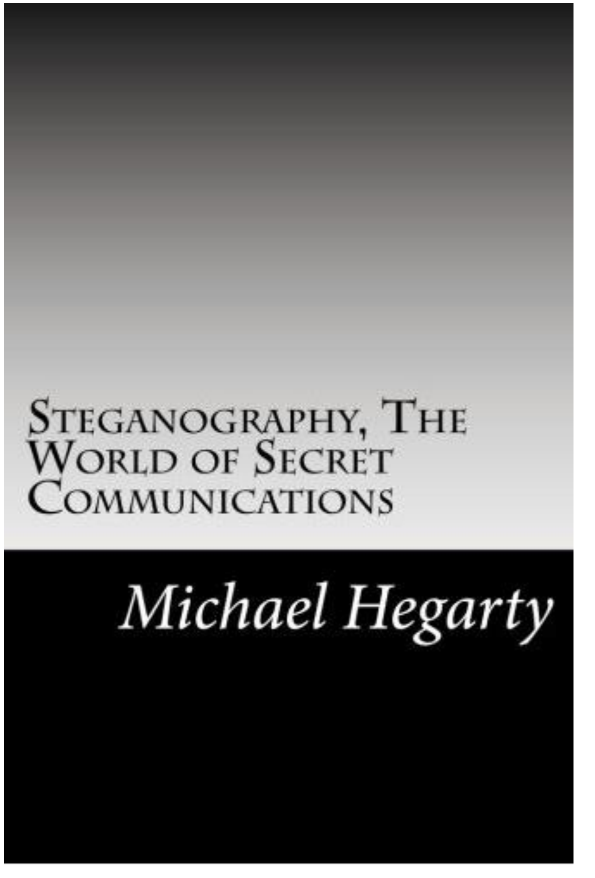
Recommended books







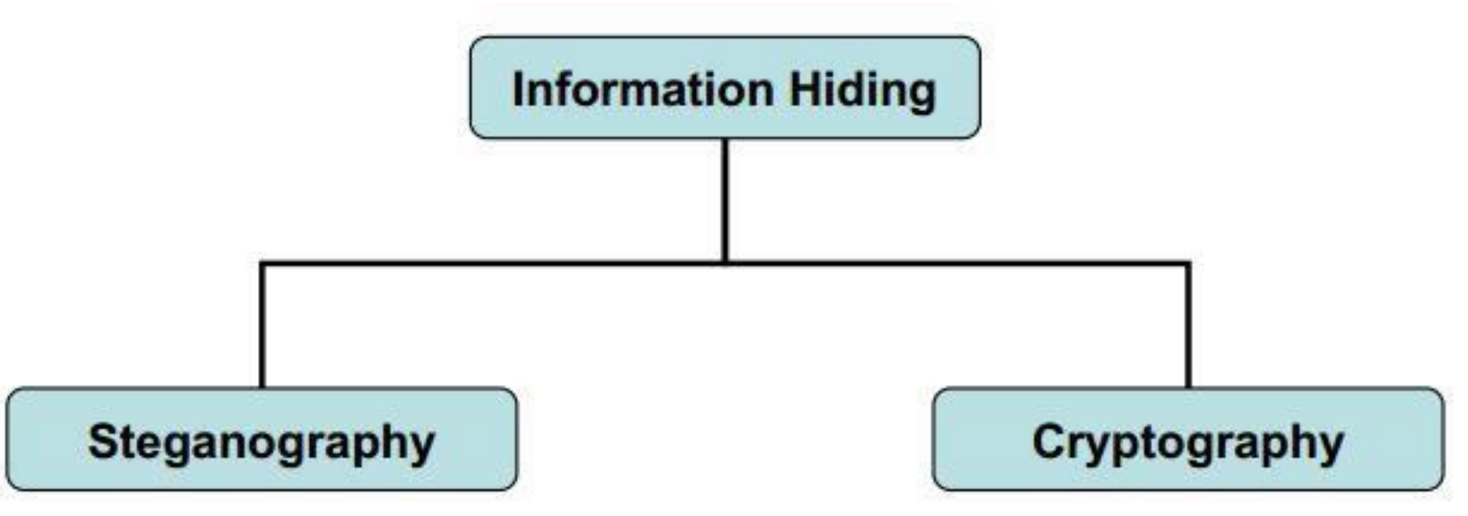
Agenda

- Introduction to steganography and steganalysis
- Machine learning in steganography and steganalysis

Introduction to steganography and steganalysis

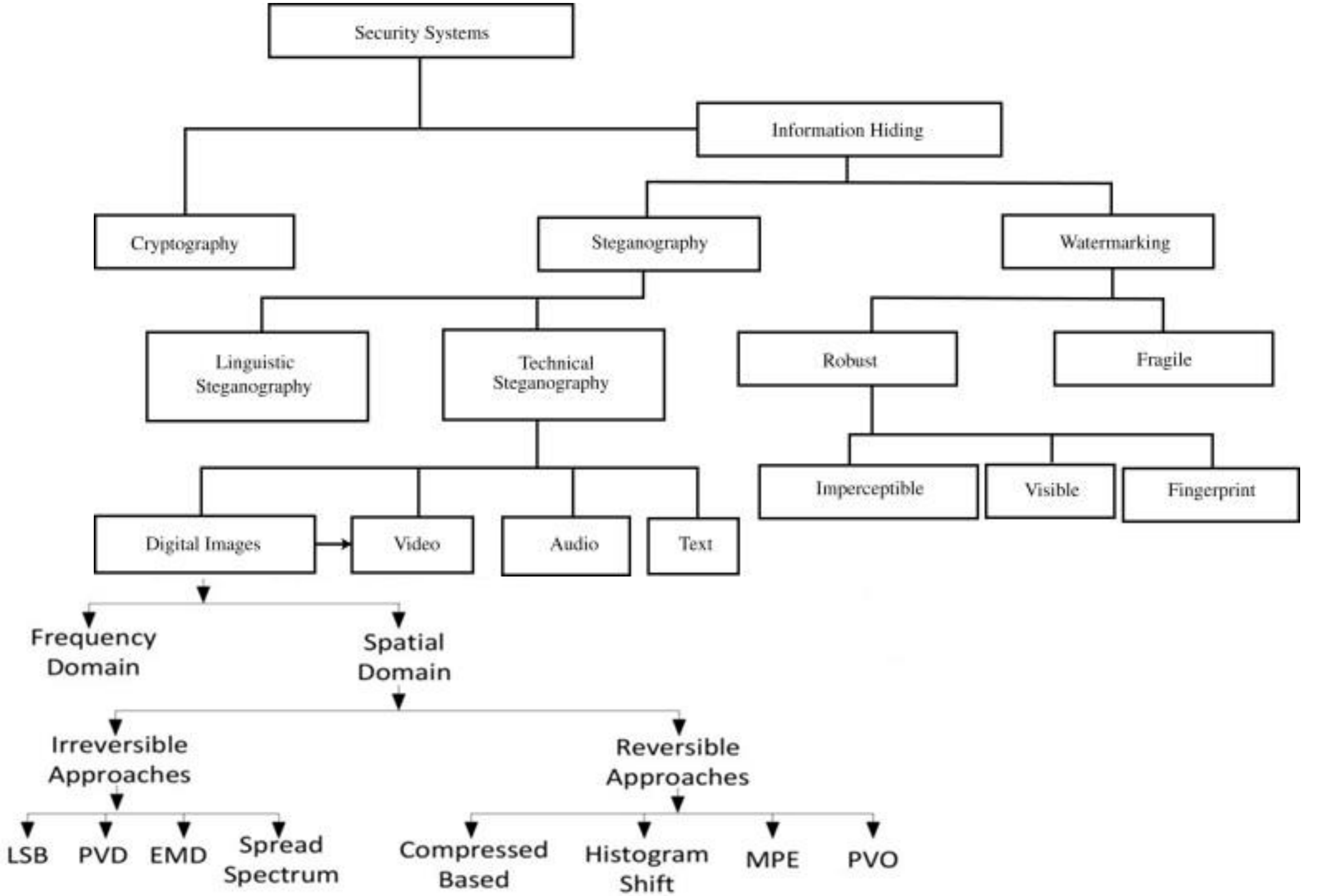
Methods of information hiding

In computer science, information hiding is the principle of segregation of the design decisions in a computer program that are most likely to change, thus protecting other parts of the program from extensive modification if the design decision is changed



Introduction to steganography and steganalysis

Methods of information hiding



Introduction to steganography and steganalysis

Steganography: hidden communication

Steganography is the art/science of communicating hiding the existence of the communication

In contrast to cryptography, where the enemy is allowed to intercept and modify messages without being able to violate the security ensured by a cryptosystem, **the goal of steganography is to hide messages inside other harmless messages** in a way that does not allow the enemy to even detect the presence of the embedded secret message

Introduction to steganography and steganalysis

Cryptography

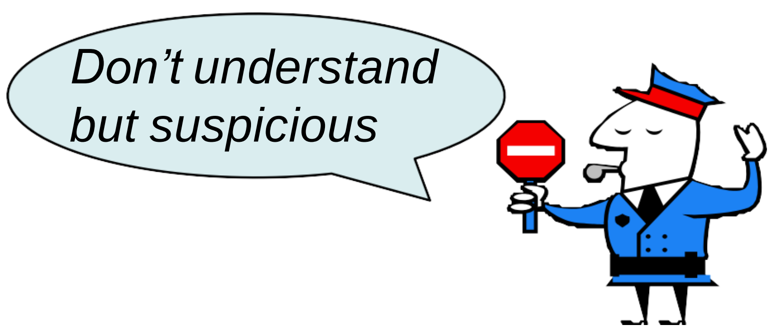
Cryptography is the study of secure communications techniques that allow only the sender and intended recipient of a message to view its contents. The term is derived from the Greek word kryptos, which means hidden. It is closely associated to encryption, which is the act of scrambling ordinary text into what is known as ciphertext and then back again upon arrival

Introduction to steganography and steganalysis

Cryptography

Key: s=v, i=e, t=y …

Bob
See you
at 8 p.m.
→ encryption → *fjwemczolamc zkhmsdjwe* → decryption →
Alice
See you
at 8 p.m.



In some cases the very existence of a message is enough to raise a suspect

Introduction to steganography and steganalysis

Steganography

- Greek Words : STEGANOS – “Covered” GRAPHIE – “Writing”
- **Steganography** is the art and science of writing hidden messages in such a way that no one apart from the intended recipient knows of the existence of the message.
- This can be achieve by concealing the existence of **information** within seemingly harmless **carriers** or **cover**
- **Carrier**: text, image, video, audio, etc.

The primary goal of steganography is to **hide a message inside another message** in a way that avoids drawing suspicion to the transmission of the hidden message. If suspicion is raised, then the goal is defeated.

Introduction to steganography and steganalysis

Steganography

Key: 6-3, 1-2, 2-2, 0, 4-5, 2-5, 11-1

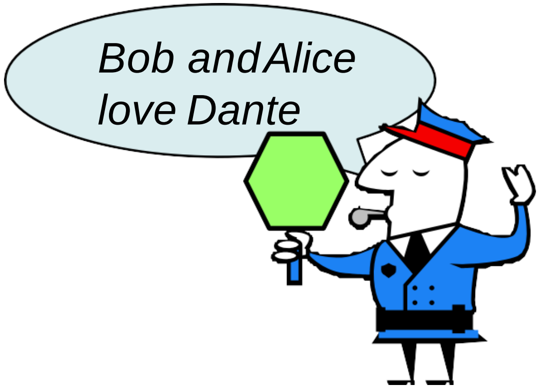
Bob
See you at 8 p.m.

Steg

In the middle of the journey of our life
I found myself in a dark forest for the straight way was lost

Decode

Alice
See you at 8 p.m.



Steganography hides the presence of the message into an innocuous host

Introduction to steganography and steganalysis

Steganography V/S Cryptography

Steganography	Cryptography
Unknown message passing	Known message passing
Steganography prevents discovery of the very existence of communication	Encryption prevents an unauthorized party from discovering the contents of a communication
Little known technology	Common technology
Technology still being develop for certain formats	Most of algorithm known by all
Once detected message is known	Strong current algorithm are resistant to attacks ,larger expensive computing power is required for cracking
Steganography does not alter the structure of the secret message	Cryptography alter the structure of the secret message

Introduction to steganography and steganalysis

Historical notes

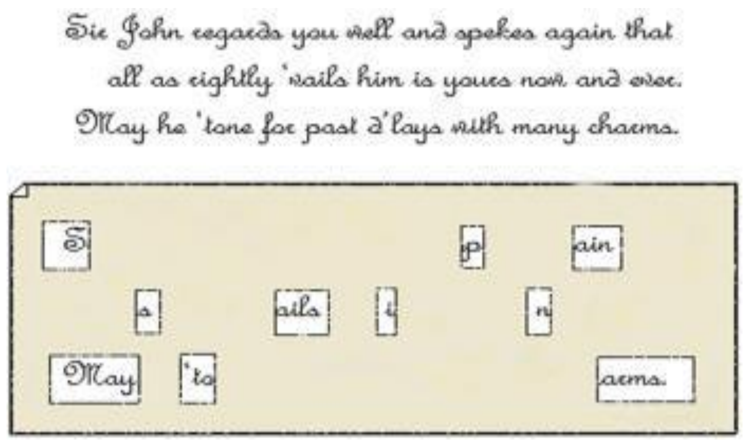
- Steganography is as old as the humans
- Ancient Chinese
- Herodotus:
  - Tatooing the head of a shaved slave
  - Writing on wood tablets then covered by wax
- Boccaccio: *Amorosa visione* (acrostic)
- Music-stego
- Invisible ink
- Modern publishing

Introduction to steganography and steganalysis

Ancient Chinese

Wax balls: the messages were written on silk and encased in balls of wax. The wax ball could then be hidden in the messenger

Paper masks: The sender and the receiver shared copies of a paper mask with a number of holes cut at random locations



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Herodotus

- Shaved slaves: messages were written over slaves heads
- Wax tablet: method was to engrave a message in a block of wood, then cover it with wax, so it looked like a blank wax tablet. When they wanted to retrieve the message, they would simply melt off the wax

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Acrostic

"My friend Bob: Until yesterday I was using binoculars for stargazing. Today I decided to try my new telescope. The galaxies in Leo and Ursa Major were unbelievable! Next, I plan to check out some nebulas and then prepare to take a few snapshots of the new comet. Although I am satisfied with the telescope, I think I need to purchase light pollution filters to block the xenon lights from a nearby highway to improve the quality of my pictures. Cheers, Alice."

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Acrostic

“My friend Bob: Until yesterday I was using binoculars for stargazing. Today I decided to try my new telescope. The galaxies in Leo and Ursa Major were unbelievable! Next, I plan to check out some nebulas and then prepare to take a few snapshots of the new comet. Although I am satisfied with the telescope, I think I need to purchase light pollution filters to block the xenon lights from a nearby highway to improve the quality of my pictures. Cheers, Alice.”

Take initial letters:

mfbuyiwubfstidttmnttgilaumwuniptcosnatpttafsotncaiaswttitintplpftbtxlfan htitqompca

Filter with p = 3.141592653689793…-> buubdlupnpsspx

Take the previous letter in the alphabet: ATTACK TOMORROW

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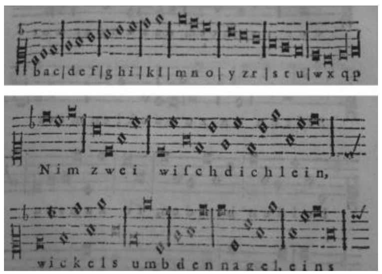
Music-stego



Gaspar Schott (17th century):
music notes are coding letters

J. S. Bach (17th-18th century):
embedded his name in the organ chorale "Vor deinen Thron" using the rule: *if the i-th note of the scale occurs k times, then the k-th letter of the alphabet is to be entered in the i-th place*

Introduction to steganography and steganalysis

Invisible inks

Lemon, urine: after burned released carbon shows up

Refined with chemistry: salt ammoniac dissolved in water

Refined with biology: some natural unique responses

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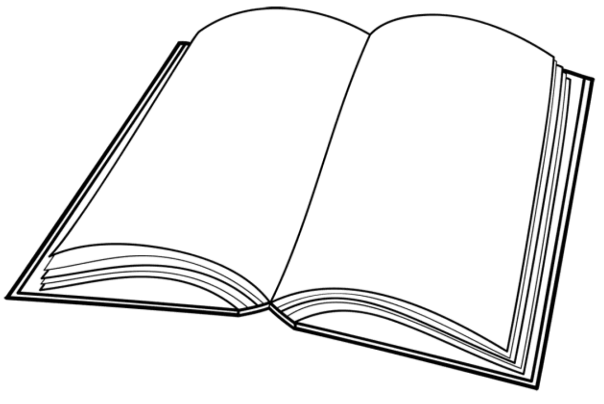
Modern publishing



Intended gaps:
False intended data

Microdots:
imperceptible dots

Line spacing:
modern publishing

Introduction to steganography and steganalysis

Modern publishing: word shifting

sentence 1:

We explore new steganographic and cryptographic algorithms and techniques throughout the world to produce wide variety and security in the electronic web called the Internet.

sentence 2:

We explore new steganographic and cryptographic algorithms and techniques throughout the world to produce wide variety and security in the electronic web called the Internet.

Introduction to steganography and steganalysis

Modern publishing: word shifting

By overlapping S1 and S2, the following sentence results

We ***explore*** *new steganographic and cryptographic algorithms and techniques throughout* ***the world*** *to produce* ***wide*** *variety and security in the electronic* ***web*** *called the Internet.*

explore the world wide web

Introduction to steganography and steganalysis

Steganography in the digital age

- Renewed interest starting from nineties
- Enabling technologies:
  - Wide band communication channels
  - Diffusion of multimedia contents
  - Possibility of using automated steganographic techniques with high payloads
- Motivations
  - Espionage, terrorism
  - Dissidents, freedom of expressing own opinions against censorship
  - Privacy protection – avoid *big-brother* scenarios

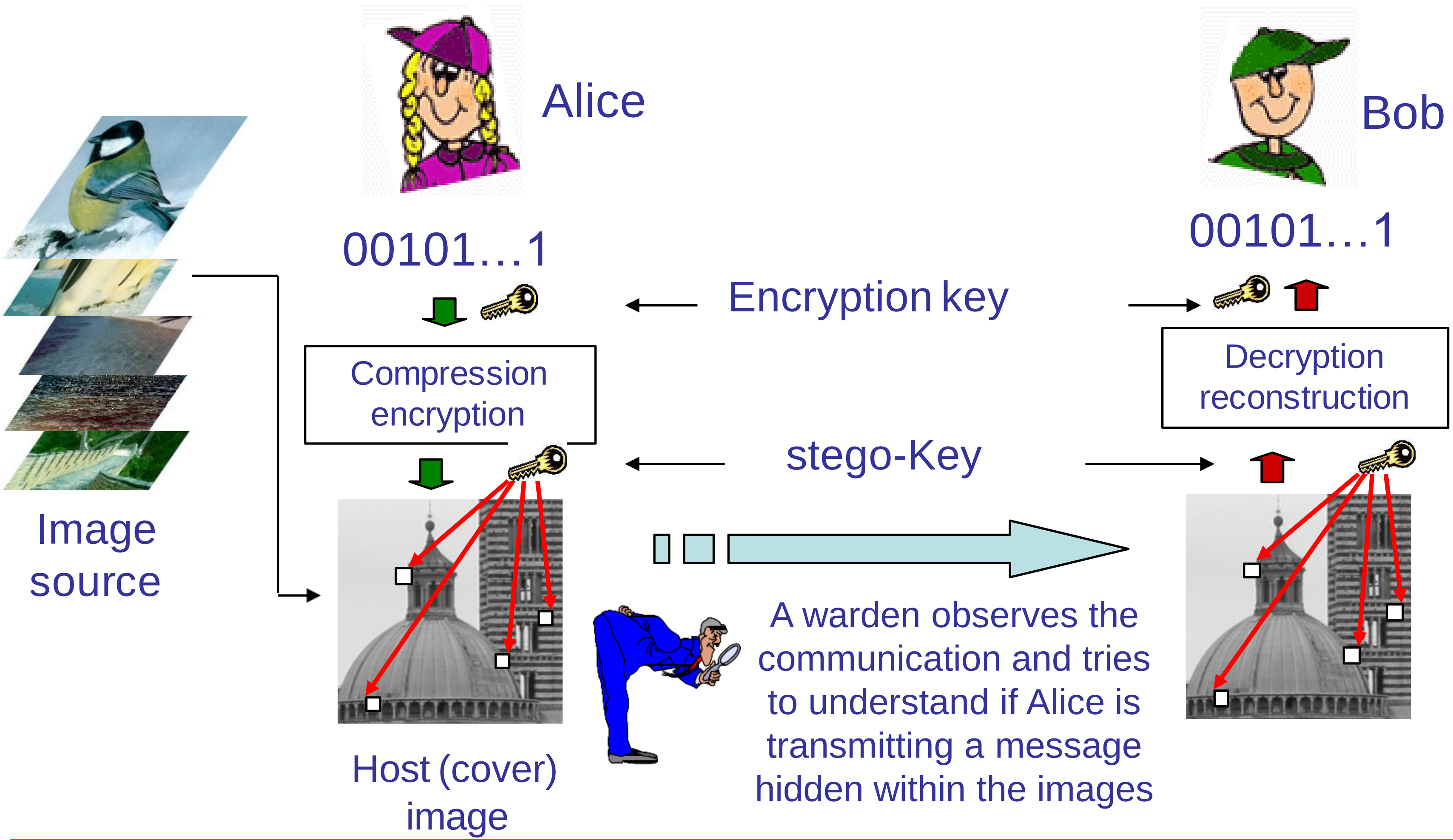
Introduction to steganography and steganalysis

Steganalysis

- Complementary motivations pushed researchers to study steganalysis
  - Techniques to reveal the presence of hidden messages (possibly without decyphering them)
- Motivations
  - Intelligence, police
  - Control of public opinion
- Regardless of motivations, the study of steganalysis is necessary to determine the security of steganographic techniques

Introduction to steganography and steganalysis

A rigorous framework: the prisoner problem



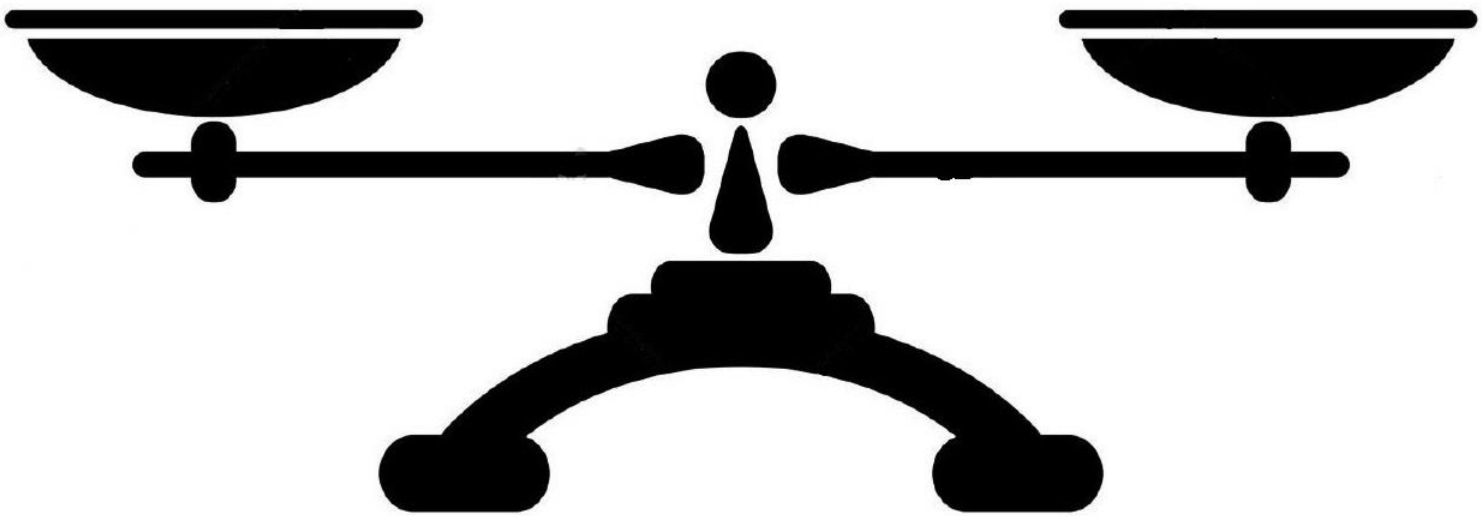
Introduction to steganography and steganalysis

Opposite requirements

In steganography designers must face with 2 opposite requirements

Invisibility
(statistical)

Capacity
(payload)



Perceptual invisibility

The hidden message must remain invisible even after the applications of signal processing techniques

message

Cover image

LSB of the green channel (original)

LSB of the green channel (stego-image)

The invisibility requirement

- Together with perceptual invisibility we require
  - **statistical invisibility**
- Assumptions on warden behavior
  - Active, **passive**
  - **Kerckhoff's principle**:
    - The warden knows the steganographic algorithm
    - The warden knows the statistics of the image source used by Alice
- Invisibility alone is not sufficient
  - Real life is always more complex than mathematical models (as cryptographers learnt quite soon)

Introduction to steganography and steganalysis

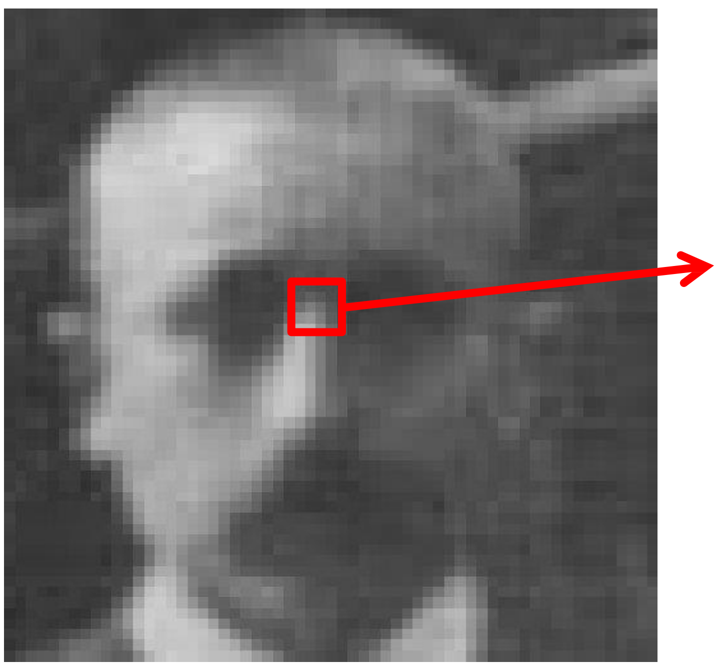
A first choice: hiding domain

- Spatial/pixel domain steganography
  - Easy to use
  - High capacity
  - Simple analysis of perceptual visibility
- Transform/compressed domain steganography (JPEG)
  - The message is conveyed by (block) DCT coefficients
  - Wide diffusion of JPEG images
  - Lower security (due to the availability of good statistical models to describe DCT coefficients)
  - Example: F5, OutGuess, Jsteg (most of them are available on the internet)

Introduction to steganography and steganalysis

Spatial/pixel domain

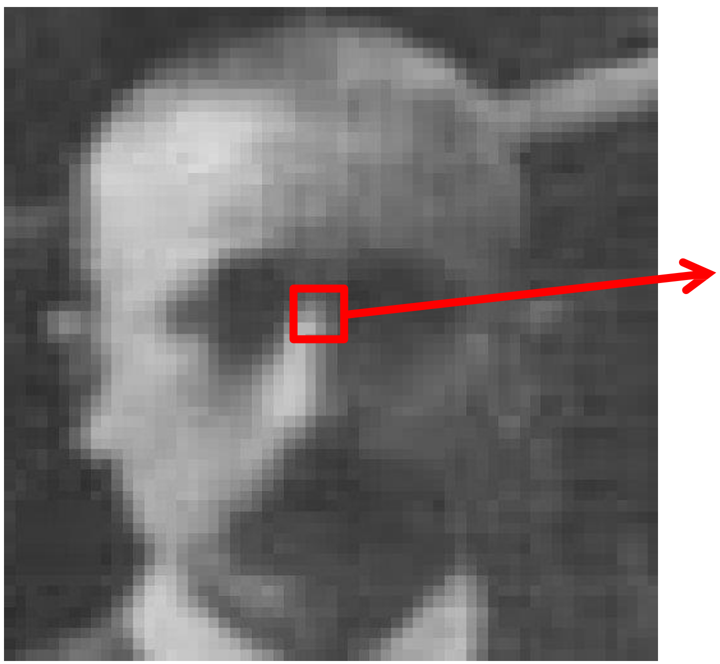
The stego-message is hidden in the array of integer numbers a digital image consists of



100	102	104	156	157	190	201	201
100	102	130	120	123	191	199	199
103	105	127	118	125	190	190	188
110	112	112	116	123	131	190	189
101	102	106	102	120	130	191	199
101	104	107	109	134	135	199	220

Transform/frequency domain

In some cases, for instance with JPEG images, the stego message is hidden into the (block) DCT coefficents of the images



-16	90	37	-17	-1	-2	-2	-1
63	10	-46	-14	12	0	0	2
-2	-9	-5	12	4	-5	-2	1
1	-3	-2	0	-3	-1	1	1
0	-2	-1	-1	0	1	1	-1
0	0	0	0	-1	0	0	0
0	-1	0	0	0	0	0	0
0	-1	0	1	0	0	0	0

Introduction to steganography and steganalysis

Three classes of steganographic algorithms

- Steganography by cover selection
- Steganography by cover synthesis
- Steganography by cover modification

Introduction to steganography and steganalysis

Steganography by cover selection

- Alice has a database of images, wherein she chooses the image corresponding to the correct message. The message can be linked to
  - Semantic image content
  - Value of a selected subset of LSB’s
  - Image (or subimage) hash
- Pros
  - Almost perfect security
- Cons
  - Very low payload
  - Example: an 8 character message (64 bit) requires a database with at least 2^{64} (10^{19}) images

Introduction to steganography and steganalysis

Steganography by cover synthesis

- Alice creates an image on-the-fly conveying the to-be-transmitted message
- Creating a realistic image is not easy. Alice could proceed as follows
  - Alice gathers several shots of the same scene
  - Alice divides the images into blocks. Each block is associated to some message bits (e.g. through a subset of LSB’s)
  - Alice builds the final image by properly assembling the blocks from various images
- Pros: good security (problems at block borders)
- Cons: still low payload (too many images needed)

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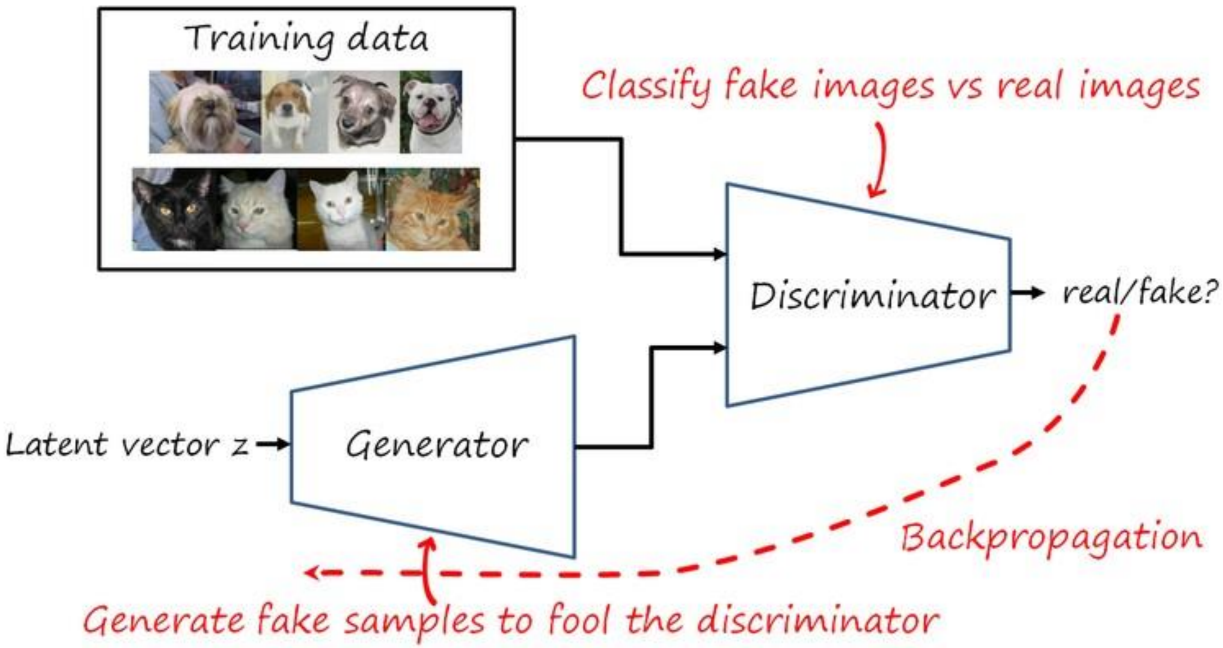
Cover synthesis by AI

- GANs and other generative models proved to be able to generate visually plausible fakes
- Two CNNs struggling following a Game-theoretic formulation



Cover synthesis by AI: examples

Images generated by GANs can be extremely realistic

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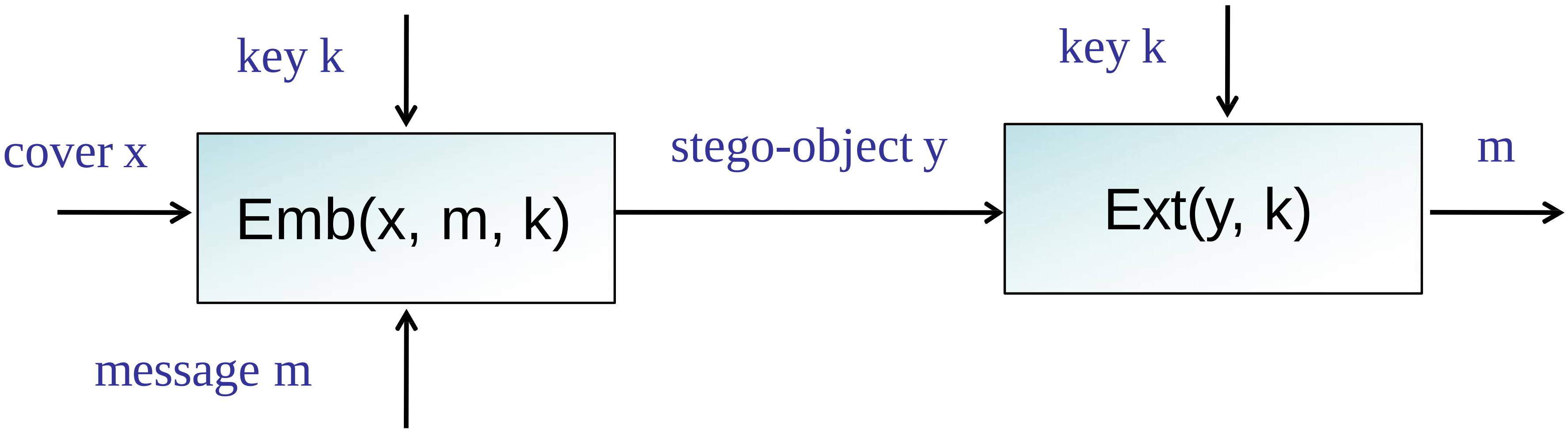
Steganography by cover modification

- By far the most common approach
- It allows large payloads, but security must be studied carefully



$$\text{Payload} = \log_2(|M|) / \text{sizeof}(x);$$

$$\text{Distortion} = d(x,y) = \sum_{i=1}^{n} \left(x(i) - y(i) \right)^2$$

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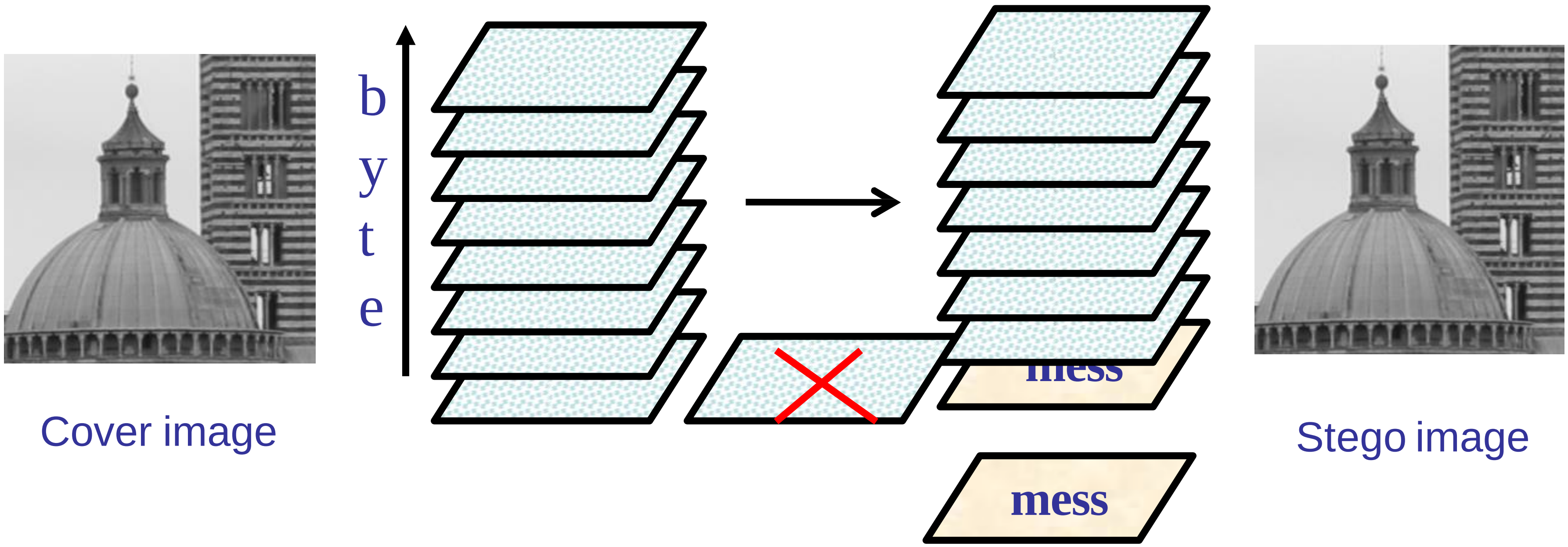
A detailed example: LSB embedding

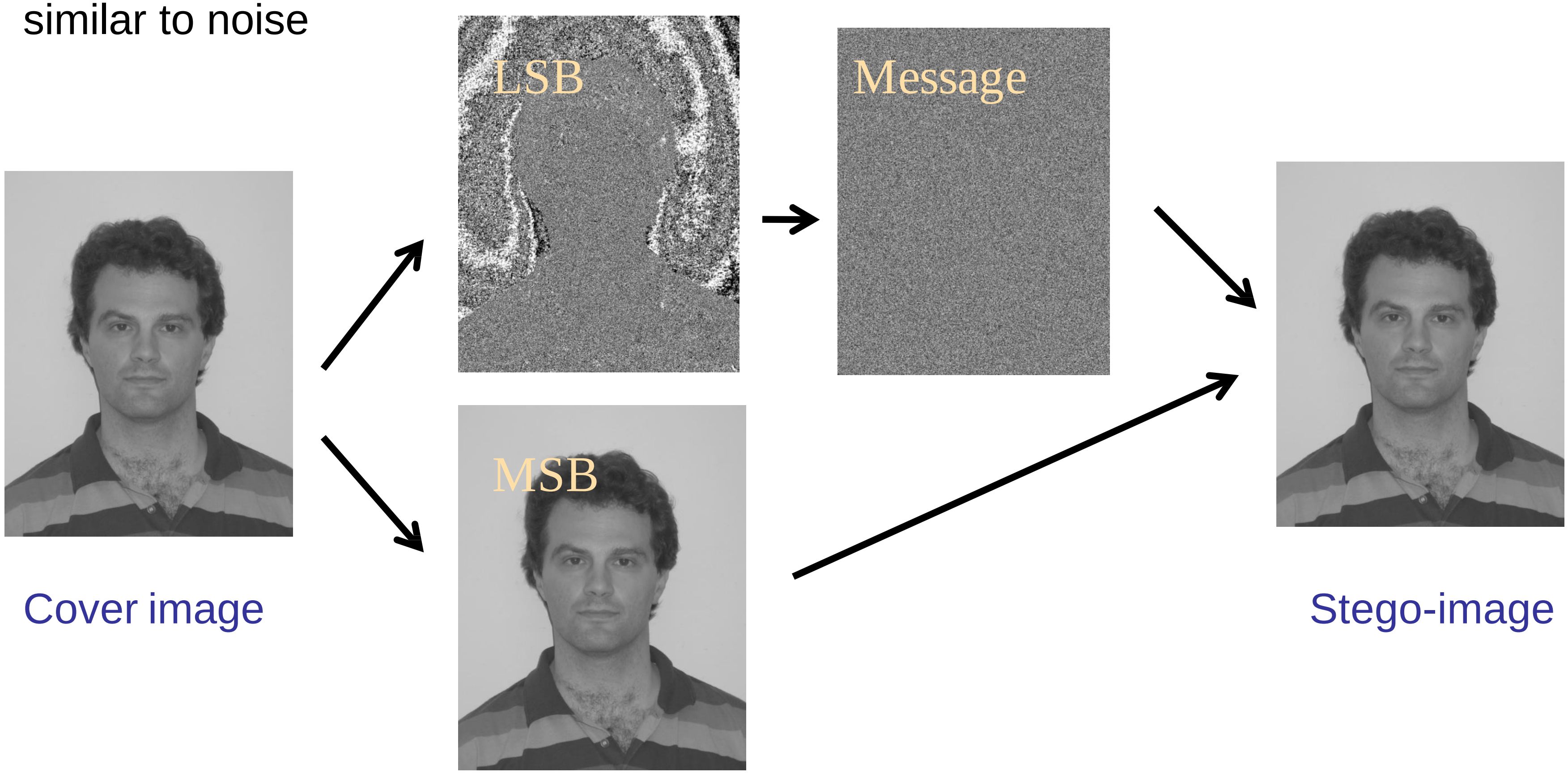
The least sensitive bits (LSB's) of the pixels of an image (or the DCT coefficients) are replaced with the stego-message (payload = 1bpp)



Introduction to steganography and steganalysis

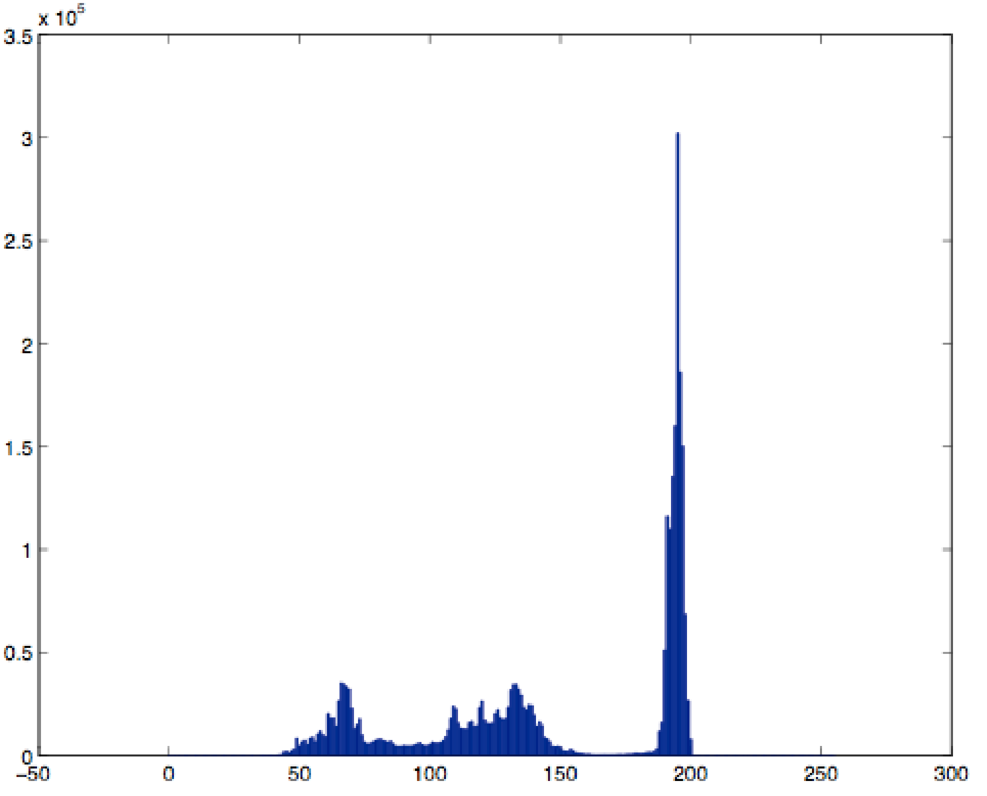
Visual imperceptibility

LSB replacement looks perfect (but is not): the LSB plane of an image is very similar to noise



Attacking LSB replacement

As a matter of fact, steganalysis of LSB replacement steganography is quite easy (at least for high payload)



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Attacking LSB replacement

- If x(i) is even we have 01100000 which remains as is or is increased by 1 -> 01100001
- If x(i) is odd we have 01100001 which remains as is or is decreased by 1 -> 01100000
- Consider the pair (0,1): (00000000, 00000001)
- Half of the pixels equal to 0 pass to 1 and half of the pixels equal to 1 pass to 0
- At the end we have about the same number of pixels = 0 and pixels = 1, that is $h_{stego}(0) = h_{stego}(1)$

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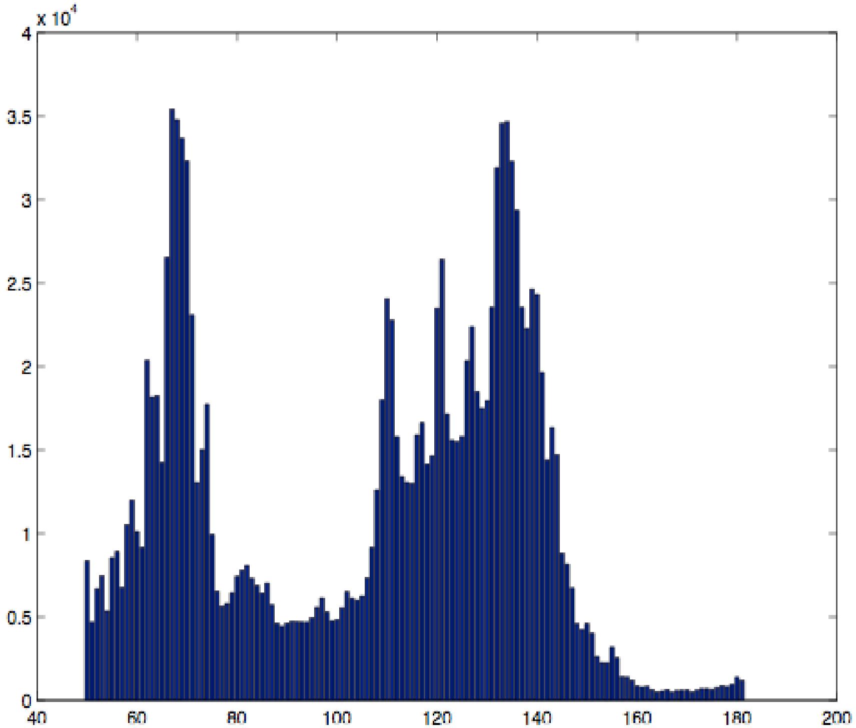
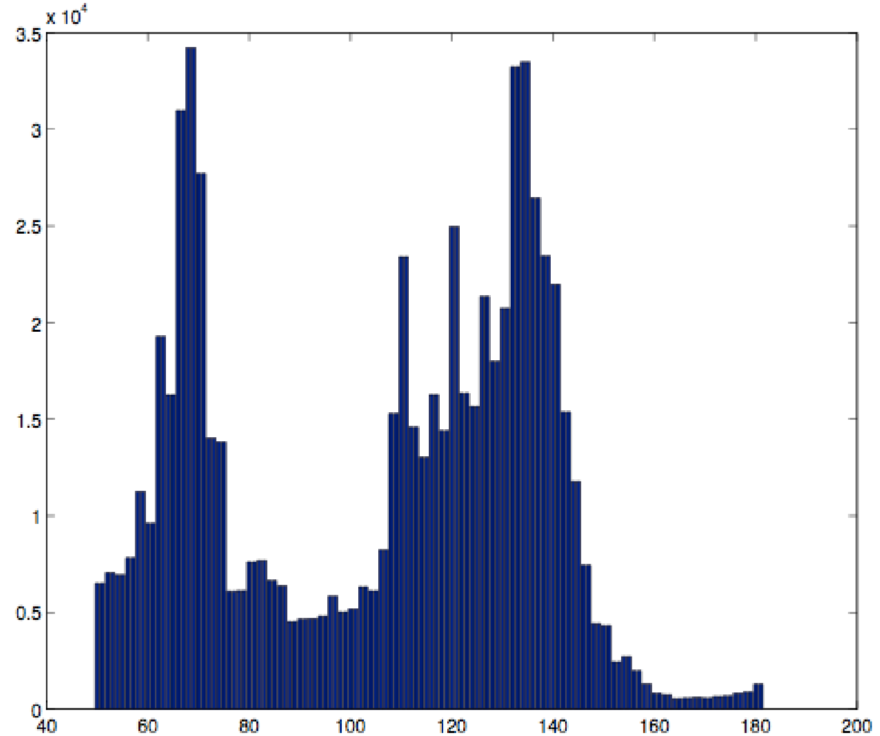
Attacking LSB replacement

The histogram of stego-images has a very characteristic behavior



Original histogram



Histogram of stego-image

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Countermeasures

- Perfect steganography requires that **all** image statistics are preserved, however
  - It is impossible to derive adequate statistical models of images (slightly better in the DCT domain)
  - It would be too complicated
- Four empirical approaches are used in practice
  - Model-preserving staganography
  - Stochastic modulation
  - Steganalysis-aware steganography
  - Distortion minimization

Introduction to steganography and steganalysis

Model-based steganography

- A model is identified to describe the image source
- Steganography acts in such a way not to modify the model
- Example: statistic restoration
  - The message is inserted in a subset of pixels (or coefficients)
  - Other pixels are modified so to restore the statistical model, e.g. the histogram
  - OutGuess -> works in this way in the DCT domain
- Nearly perfect security as long as the steganalysis relies only on the adopted statistical model (in practice this is never the case)

Stochastic modulation

- It simulates the noise added to the image during the acquisition phase
- Steganography works by adding a noise that resembles acquisition noise
  - Thermal noise
  - Quantization noise
  - PRNU
- It allows rather high payloads (0.8 bpp)

Introduction to steganography and steganalysis

Steganalysis-aware steganography

- The steganographer acts in such a way to eliminate (reduce) the artefacts exploited by the steganalyzer
- Example: ±1-steg
  - If the LSB is the correct one doesn't do anything
  - If the LSB is wrong, add or subtract 1 randomly
  - Observation: ±1steg does not modify only the LSB

    01111111+1=10000000
- Security increases dramatically since the histogram does not change significantly (convolution)
- F5 uses ±1-steg in the frequency domain

Distortion (impact) minimization

- Most modern approach
- Define a cost function
  - How much does it cost to modify a certain pixel? Say ρ(i)
  - Overall cost = $\sum_{i=1}^{n} \rho(i)[x(i) - y(i)]^2$
- Identify an embedding rule which minimizes the embedding cost
  - F5 is optimum from this point of view (DCT domain)

Introduction to steganography and steganalysis

Typical payloads

- Payload
  - from 0.1 to 0.5 bpp in the pixel domain:

    1000x1000 image => ~ 40Kbyte
  - up to 0.8 bpnzc in the DCT domain:

    The actual payload depends on the image content. A realistic value is around 20Kbyte for a 1000x1000 image

Introduction to steganography and steganalysis

Steganalysis

The application scenario is of the outmost importance together with the information available to the warden

- Blind vs targeted steganalysis
- Knowledge of cover image statistics
- Knowledge of payload

Steganalysis = hypothesis test

- Rigorous formulation
- Observables: $\mathbf{y} = \{y_1, y_2 \ldots y_N\}$
  - Image pixels, audio signal samples, etc.
  - Often to simplify the problem the analysis relies on some functions of y (features)
- Two alternative hypothesis
  - H_0 : **y** does not contain a hidden message
  - H_1 : **y** contains a hidden message
- Optimum decision with respect to a certain criterion

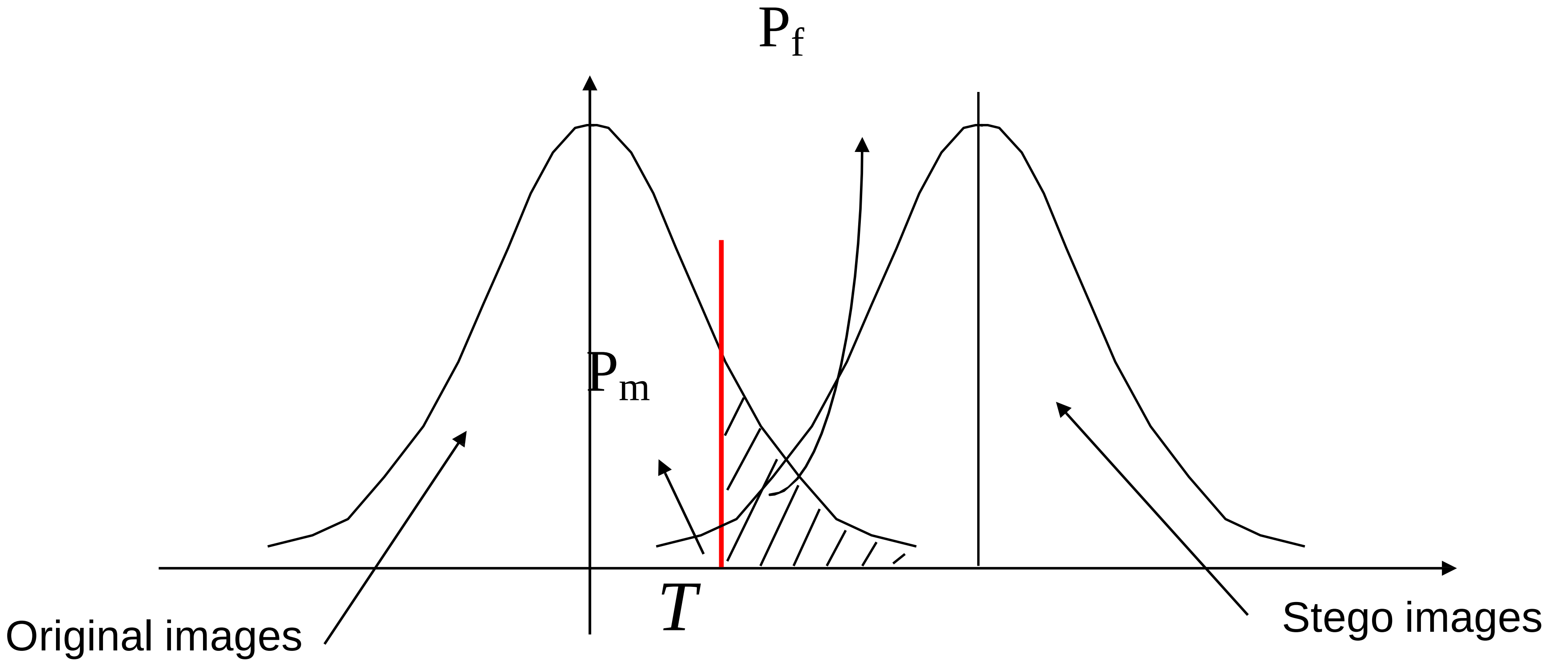
Introduction to steganography and steganalysis

Steganalysis = hypothesis test

- Bayes criterion
  - Minimization of overall error probability
  - Difficult to apply since a prior probabilities are not known
- Neyman-Pearson criterion
  - False alarm probability
    - Decide in favour of H_1 when H_0 holds
  - Missed detection probability
    - Decide in favour of H_0 when H_1 holds
  - N-P: minimize P_m for a given (maximum) P_f
- In steganalysis we must first fix P_f and then decide how to use the result of the test

Example

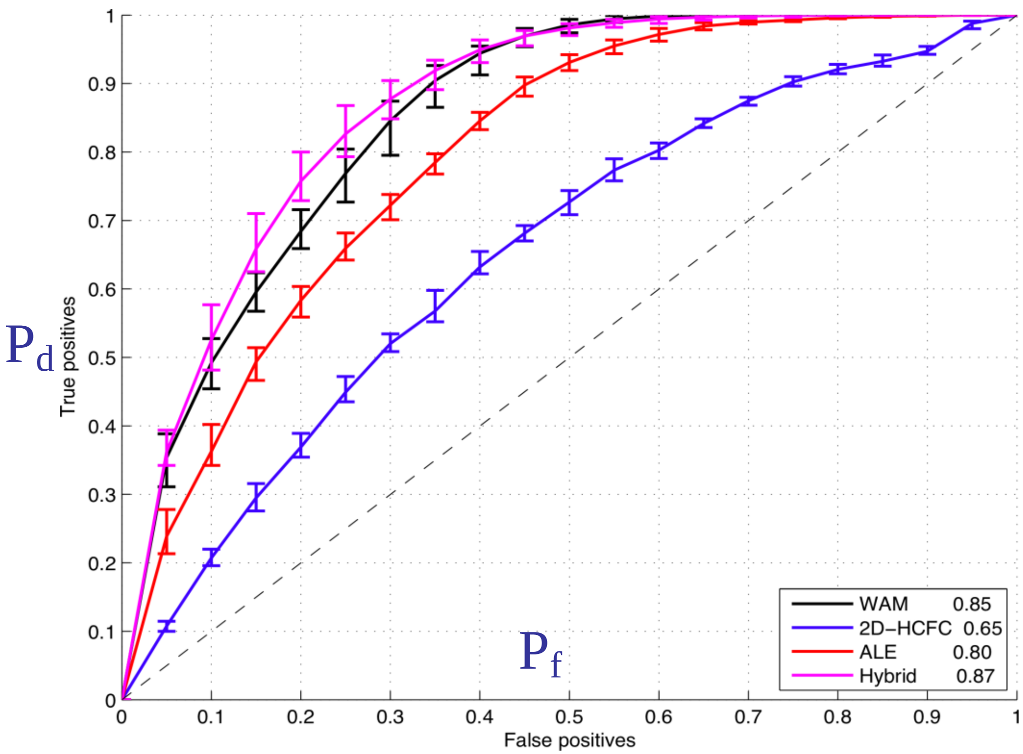
Let us assume that the test relies on a single statistics with known pmf (Gaussian) under both H_0 and H_1



Introduction to steganography and steganalysis

ROC curve

For any value of P_f (threshold) we find a P_m. The plot showing $P_d = 1 - P_m$ as a function of P_f is called ROC curve



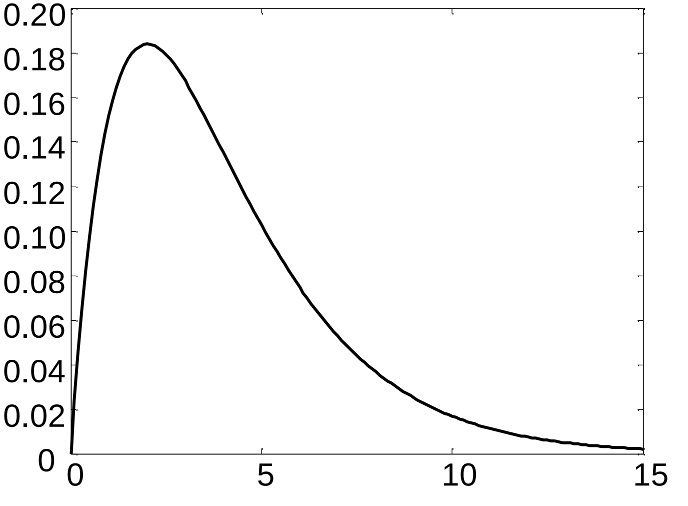
The goodness of a steganalyzer is evaluated by means of the ROC curve or its AUC. Perfect security requires that performance are equal obtained by means of a random guess (diagonal ROC, AUC = ½) .

Introduction to steganography and steganalysis

Example

- Let us assume that the pdf of the image source is known. In this case the steganalyzer can use a Chi Suare test
- Divide the pdf in several intervals (bins)
- Compute how many times the observed samples fall in the bins: let us indicate this value as n_i
- Given the pdf let indicate with p_i the probability that a sample falls into the i-th bin

$$\chi^2 = \sum_{i=1}^{n} \frac{(n_i - np_i)^2}{np_i}$$



- High values are taken as an evidence in favour of H_1

Introduction to steganography and steganalysis

Choice of statistics (feature)

- In targeted seteganalysis we use few ad-hoc statistics
- Example: LSB replacement steganalysis

Given an image and its histogram, we can use a Chi-square test in which the assumed pmf is

$$h_{Hp1}(2k) = h_{Hp1}(2k+1) = \frac{h(2k)+h(2k+1)}{2}$$

Such a test reveals the presence of LSB steganography (at 1 bpp) with great accuracy. Steganalysis is obviously more difficult at low payloads

Introduction to steganography and steganalysis

Choice of statistics (feature)

- With blind steganalysis everything is more difficult
- If source statistics are known we can still use targeted features
- Otherwise
  - Compute many features (> 100) that do not depend on image content
  - Train a classifier with properly chosen examples
    - Neural networks, Support Vector Machines (SVM)
- ROC curves are evaluated empirically on a test set
- CNN applied directly in the pixel domain are rapidly replacing SVMs

Introduction to steganography and steganalysis

Summary

- Several steganographic techniques exist with a large number of available software packages
  - Security looks trivial but is not
  - Need to know at least basic principles
  - Take care of system attacks
- Steganalysis
  - Reliable in some selected cases, but difficult in general
  - Strongly dependent on application scenario
- Work in progress

Conclusions and references

- Jessica Fridrich (Binghamton University, USA):
  http://www.ws.binghamton.edu/fridrich
- Min Wu (University of Maryland, USA):
  https://user.eng.umd.edu/~minwu
- Mauro Barni (University of Siena, Italy):
  https://scholar.google.it/citations?user=ntRScY8AAAAJ&hl=en
- Nasir Memon (New York University, USA):
  https://engineering.nyu.edu/faculty/nasir-memon